5 Fig. 5.1 shows the structure of a geophone which is used by geophysicists to determine the speed of seismic waves traveling within the ground layer.

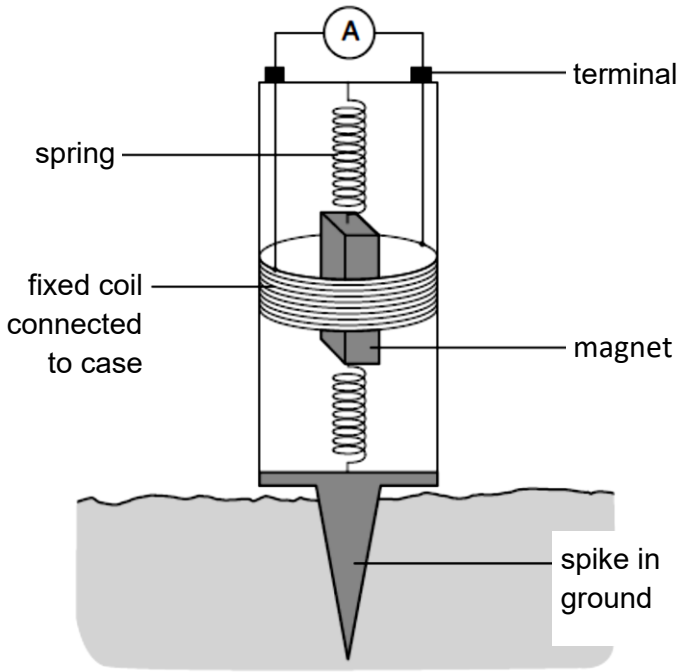

Fig. 5.1

The spike of the geophone is inserted into the ground. When a seismic vibration moves the case and coil, the magnet remains stationary due to its inertia.

(a) State Faraday's law of electromagnetic induction.

.....

..... [1]

(b) Using Faraday's law, explain how an e.m.f. is generated between the terminals when seismic waves pass through the ground.

.....

.....

.....

..... [2]

(c) A coil of 50 turns generates a maximum e.m.f. of 85 mV in a geophone.

Calculate the rate of change of magnetic flux needed to generate this e.m.f.

rate of change of magnetic flux = …… Wb s^{-1} [2]

(d) Suggest 2 changes to the geophone which will make it more sensitive to the seismic vibrations.

1. ..... [1]

2. ..... [1]